Operations Runbook

Standard Operating Procedures for Deployment, Incidents & Recovery

DevOps & SRE Team | Version 2.3

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1. Deployment Standard Operating Procedure

All production deployments follow the Blue-Green deployment model orchestrated via GitHub Actions and ArgoCD. Deployments are only permitted during the deployment window: Monday through Thursday, 10:00 AM – 4:00 PM ICT. Friday deployments require VP Engineering approval. No deployments are allowed during code freeze periods or within 48 hours of a major client demo.

1.1 Pre-Deployment Checklist

Step	Action	Responsible	Tool/Command	Pass Criteria
1	All PRs merged and approved (min 2 reviewers)	Tech Lead	GitHub — check PR status	0 open PRs targeting release branch
2	All CI checks passing (unit, integration, SAST, DAST)	CI/CD Pipeline	GitHub Actions	All checks green, 0 failures
3	Staging environment tested and signed off	QA Lead	TestRail — test run report	100% P1/P2 tests pass, <2% P3 failures
4	Database migration reviewed by DBA	DBA (Tran Duc)	pg_dump diff + review	No destructive changes without backup plan
5	Release notes published to #releases channel	PM / Tech Lead	Slack + Confluence	All stakeholders notified
6	Rollback plan documented and tested in staging	DevOps	ArgoCD rollback dry-run	Rollback completes in <5 minutes
7	Monitoring dashboards reviewed (baseline metrics)	SRE	Datadog — service dashboard	Error rate <0.1%, latency p99 <500ms
8	Change request approved in ServiceNow	Release Manager	ServiceNow CR ticket	CAB approval or emergency approval

1.2 Deployment Execution Steps

Step	Action	Command / Procedure	Expected Duration	Rollback Trigger
1	Enable maintenance page (if needed)	kubectl apply -f maintenance.yaml	30 seconds	N/A
2	Deploy to blue environment	argocd app sync --revision	3-5 minutes	Image pull failure or CrashLoopBackOff
3	Run smoke tests against blue	pytest tests/smoke/ --env=blue	2 minutes	Any smoke test failure
4	Gradual traffic shift: 10% -> 25% -> 50% -> 100%	istio traffic-split update	15 minutes total (5min per stage)	Error rate >1% or p99 >2s at any stage

Step	Action	Command / Procedure	Expected Duration	Rollback Trigger
5	Monitor for 30 minutes (bake time)	Datadog real-time dashboard	30 minutes	Error rate >0.5% or customer-reported issue
6	Mark green environment for decommission	argocd app delete	1 minute	N/A
7	Close change request in ServiceNow	Update CR status to Completed	1 minute	N/A

1.3 Rollback Procedure

If rollback is triggered at any stage, execute these steps immediately:

Step	Action	Command	Expected Duration
1	Shift 100% traffic back to green (old version)	istio traffic-split rollback	30 seconds
2	Verify green environment is healthy	curl -s https://api.techviet.vn/health | jq .status	10 seconds
3	Delete blue (failed) deployment	argocd app delete	1 minute
4	Notify #incidents channel with rollback reason	Slack — /incident rollback-report	2 minutes
5	Create post-deployment failure ticket in Jira	Jira — type: Bug, priority: P1	5 minutes
6	Schedule post-mortem within 48 hours	Google Calendar — invite release team	5 minutes

2. Incident Response Procedure

TechViet follows a structured incident response framework based on PagerDuty's incident management best practices. All incidents are tracked in ServiceNow with Slack as the primary communication channel. The on-call rotation covers 24/7 with primary and secondary responders from each team.

2.1 Severity Classification

Severity	Definition	Response Time	Update Frequency	Escalation	Example
P1 — Critical	Complete service outage affecting all users or data loss/breach	15 minutes	Every 30 minutes	VP Eng + CTO within 30min	Production database down, API gateway unreachable, data breach detected
P2 — High	Major feature degraded, affecting >30% of users or revenue impact	30 minutes	Every 1 hour	Engineering Manager within 1hr	Payment processing failing, search returning no results, auth service degraded
P3 — Medium	Minor feature issue, workaround available, <10% of users affected	4 hours	Every 4 hours (business)	Team Lead next business day	PDF export broken, Slack integration delayed, dashboard showing stale data
P4 — Low	Cosmetic issue, minor bug, no user impact	Next business day	Weekly status	Sprint backlog	Typo in email template, UI alignment issue, non-critical log warning

2.2 On-Call Escalation Path

Time Elapsed	Action	Who	Contact Method
0 min	Alert fires — Primary on-call receives PagerDuty notification	Primary On-Call Engineer	PagerDuty push + SMS + phone
5 min	If no acknowledgment — auto-escalate to secondary	Secondary On-Call Engineer	PagerDuty push + phone call
15 min	If P1 — Incident Commander declared, war room opened	Engineering Manager (IC)	Slack #incident-{id} channel created
30 min	If P1 unresolved — escalate to VP Engineering	VP Engineering (Le Van Hung)	Phone: +84-xxx-xxx-001
1 hour	If P1 unresolved — CTO and affected client PM notified	CTO + Client Success	Phone + Email
2 hours	If P1 unresolved — CEO briefed, external comms prepared	CEO + PR Team	Executive Slack + email

2.3 Incident War Room Procedures

For P1 and P2 incidents, an incident war room is created in Slack (#incident-{YYYYMMDD}-{seq}). The Incident Commander (IC) owns the channel and coordinates all activities. Key rules: (1) Only IC makes decisions about customer communication, (2) All actions are logged in the channel with timestamps, (3) IC posts status updates at the defined frequency, (4) War room stays open until post-mortem is completed.

3. Database Emergency Procedures

Scenario	Immediate Action	Command / Tool	Recovery Time (RTO)	Data Loss (RPO)
Primary DB down	Promote read replica to primary	aws rds failover-db-cluster --db-cluster-id phoenix-prod	5 minutes	0 (synchronous replication)
Connection pool exhausted	Kill idle connections, increase pool	SELECT pg_terminate_backend(pid) WHERE state='idle' AND query_start < now()-'5min'	2 minutes	0
Disk space >90%	Vacuum, archive old data, expand volume	VACUUM FULL ; aws ec2 modify-volume --size 500	15 minutes	0
Corrupted index	Reindex concurrently	REINDEX INDEX CONCURRENTLY idx_name	5-30 minutes	0
Accidental data deletion	Restore from point-in-time backup	aws rds restore-db-cluster-to-point-in-time --restore-to-time	30 minutes	Up to 5 minutes
Full cluster failure	Restore from latest snapshot + replay WAL	aws rds restore-db-cluster-from-snapshot	1-2 hours	Up to 1 hour

4. Monitoring & Alerting Reference

Service	Dashboard URL	Key Metrics	Alert Threshold	PagerDuty Service
API Gateway	datadog.com/d/kong-overview	Request rate, error rate, latency p50/p95/p99	Error rate >1% for 5min, p99 >3s	api-gateway-prod

Service	Dashboard URL	Key Metrics	Alert Threshold	PagerDuty Service
User Service	datadog.com/d/user-svc	Auth success rate, token refresh rate, DB connections	Auth failure >5% for 3min	user-service-prod
Project Service	datadog.com/d/project-svc	API latency, Jira sync lag, DB query time	Jira sync lag >5min, query >2s	project-service-prod
PostgreSQL	datadog.com/d/postgres-prod	Connections, replication lag, disk usage, deadlocks	Replication lag >10s, disk >85%	database-prod
Kafka	datadog.com/d/kafka-prod	Consumer lag, partition count, throughput	Consumer lag >10000, broker down	kafka-prod
Kubernetes	datadog.com/d/eks-cluster	Pod restarts, CPU/memory utilization, node health	Pod restart >3 in 10min, CPU >80%	kubernetes-prod